

Table 8.9 Factor Pricing Models, 1974–2013

	Alpha*	Market	Size	Value	Momentum	Bond	R ²
Five-factor model	5.30 (2.67)	0.50 (8.32)	−0.06 (1.10)	0.08 (1.32)	0.20 (4.25)	0.37 (3.50)	0.44
Four-factor model	5.94 (2.99)	0.53 (9.64)	−0.09 (1.83)	0.09 (1.43)	0.21 (4.39)	–	0.44
Three-factor model	5.80 (3.25)	0.47 (8.29)	–	–	0.17 (5.00)	0.39 (3.88)	0.43

*Alpha is annualized. Newey-West robust t-statistics are in parentheses.

Under all three of these factor models, GEM provides economically and statistically significant risk-adjusted excess returns (alpha). Since GEM is long only, we naturally see highly significant coefficients on the stock and bond market factors. GEM also has a highly significant loading on stock momentum, showing that relative strength momentum plays a significant role in the strong performance of GEM.

SIMPLE AND EFFECTIVE

GEM is not only effective in providing high and significant risk-adjusted returns, but it is also parsimonious. This is a word used by economists to make them look smart. It means simple and straightforward. Einstein once said, “Everything should be kept as simple as possible, but no simpler.” In the words of Antoine de Saint-Exupéry, “Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away,” and in the words of Mr. Rogers, “Deep and simple is far more essential than shallow and complex.”

Highly optimized methods are often complex, fragile, and prone to failure. GEM, on the other hand, is simple and robust. It uses only U.S. equities, non-U.S. equities, and aggregate bonds. Its only parameter is a 12-month look-back period validated in hundreds of in- and out-of-sample momentum studies across many diverse markets and over two centuries of market data. These in- and out-of-sample results and the simplicity and